CSC1016 Group Project 2

Healthcare Appointment Management System

System Analysis & Design Report

1. Introduction

1.1 Group Members and Contributions

Member	Contributions	Est. %
Vasileios Papakostas	Introduction, Process Landscape, Process Design for Processes 1 & 2	25%
Danylo Dudar	Process Design for Processes 3 & 4, UML Class Diagram	25%
Ilja Solovejs	Process Design for Processes 5 & 6, System Design, Collaboration Diagrams	25%
Denys Maksymov	Process Design for Processes 7 - 9, Discussion, Assumptions & Constraints, UI Design (Bonus)	25%

1.2 Description of Client System

The client group (Group 11:11) proposed a Healthcare Appointment Management System for Irish clinics and GP practices. The system addresses a real-world problem where manual appointment management leads to scheduling conflicts, incomplete patient records, delayed confirmations, and high administrative workload.

The client identified five core processes:

- Appointment Request Submission: patients submit requests online or by phone
- Patient Information Validation: verifying patient details and eligibility
- Appointment Scheduling and Approval: matching patient requests with available providers
- Appointment Confirmation and Communication: notifying patients of outcomes
- Appointment Fulfilment and Closure: conducting appointments and closing cases

As system analysts and designers for this project, our goal is to transform these high-level requirements into a detailed, implementation-ready process design using advanced BPMN models, UML class diagrams, and supporting system artefacts.

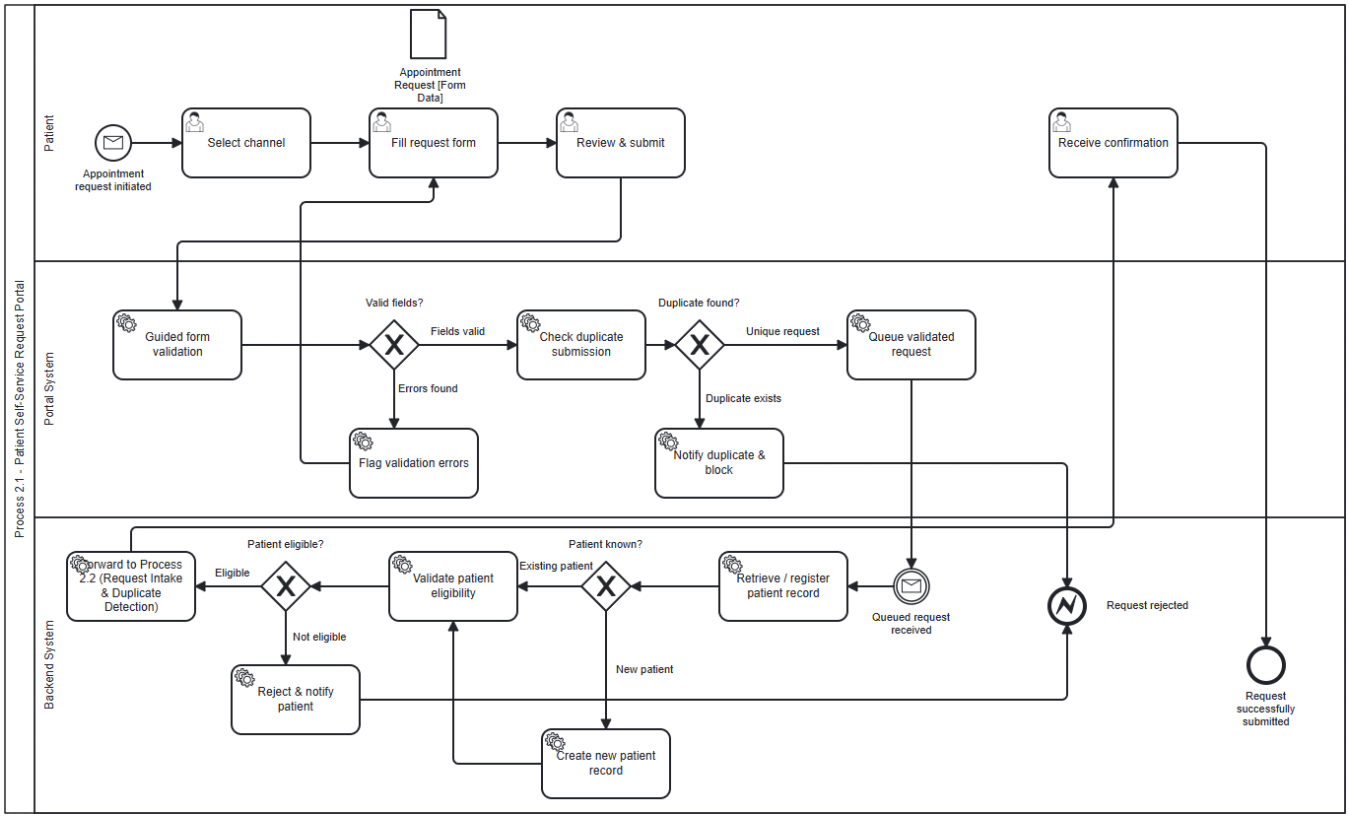
2. Process Landscape

Our design expands the original five processes into nine distinct processes. Several of the original processes contained significant complexity that warranted decomposition for clarity, maintainability, and real-world feasibility. The table below maps each designed process to its originating client process.

#	Designed Process Name	Original Client Process	Description
2.1	Patient Self-Service Request Portal	1. Appointment Request Submission	Patient-facing web/phone portal for submitting appointment requests with guided validation
2.2	Request Intake & Duplicate Detection	1. Appointment Request Submission	Back-end intake handler that checks for duplicate submissions and queues validated requests
2.3	Patient Record Retrieval & Registration	2. Patient Information Validation	Retrieves existing patient records or registers new patients in the system database
2.4	Patient Eligibility Validation	2. Patient Information Validation	Validates patient data completeness, format, and eligibility rules before proceeding
2.5	Provider Availability Check & Scheduling	3. Appointment Scheduling & Approval	Checks real-time provider availability and schedules the appointment if a slot is found
2.6	Alternative Scheduling & Patient Negotiation	3. Appointment Scheduling & Approval	Handles the case where no provider is available and offers alternatives to the patient
2.7	Appointment Confirmation & Reminder Dispatch	4. Appointment Confirmation & Communication	Generates and sends confirmation messages and schedules automated reminders
2.8	Appointment Fulfilment	5. Appointment Fulfilment & Closure	Covers patient check-in, identity verification, and appointment conduct
2.9	Post-Appointment Processing & Case Closure	5. Appointment Fulfilment & Closure	Records results, determines follow-up need, and closes or escalates the case

3. Process Design

3.1 Process 2.1 - Patient Self-Service Request Portal



Overview

This process handles the patient-facing side of appointment request submission. It extends the original Process 1 by incorporating advanced BPMN elements such as timer boundary events, error handling, and message events to reflect real-world complexity.

Key Design Improvements Over Original

- The original diagram used a basic exclusive gateway ("All fields complete?") with a loop back to the start. Our design adds an explicit validation gateway that routes incomplete submissions back to the form-fill task for correction, ensuring all required fields are populated before submission proceeds.
- A message start event replaces the generic start event, reflecting that the process is triggered by an incoming patient message (web request or phone call).
- An error end event represents the formal rejection of duplicate-blocked or ineligible requests, with a structured error code (ERR_REQUEST_REJECTED) for downstream processing.
- User tasks are distinguished from service tasks: form entry is a user task; duplicate detection is a service task.
- A data object ('Appointment Request Form') is attached to the submission task to make data flow explicit.

BPMN Elements Used

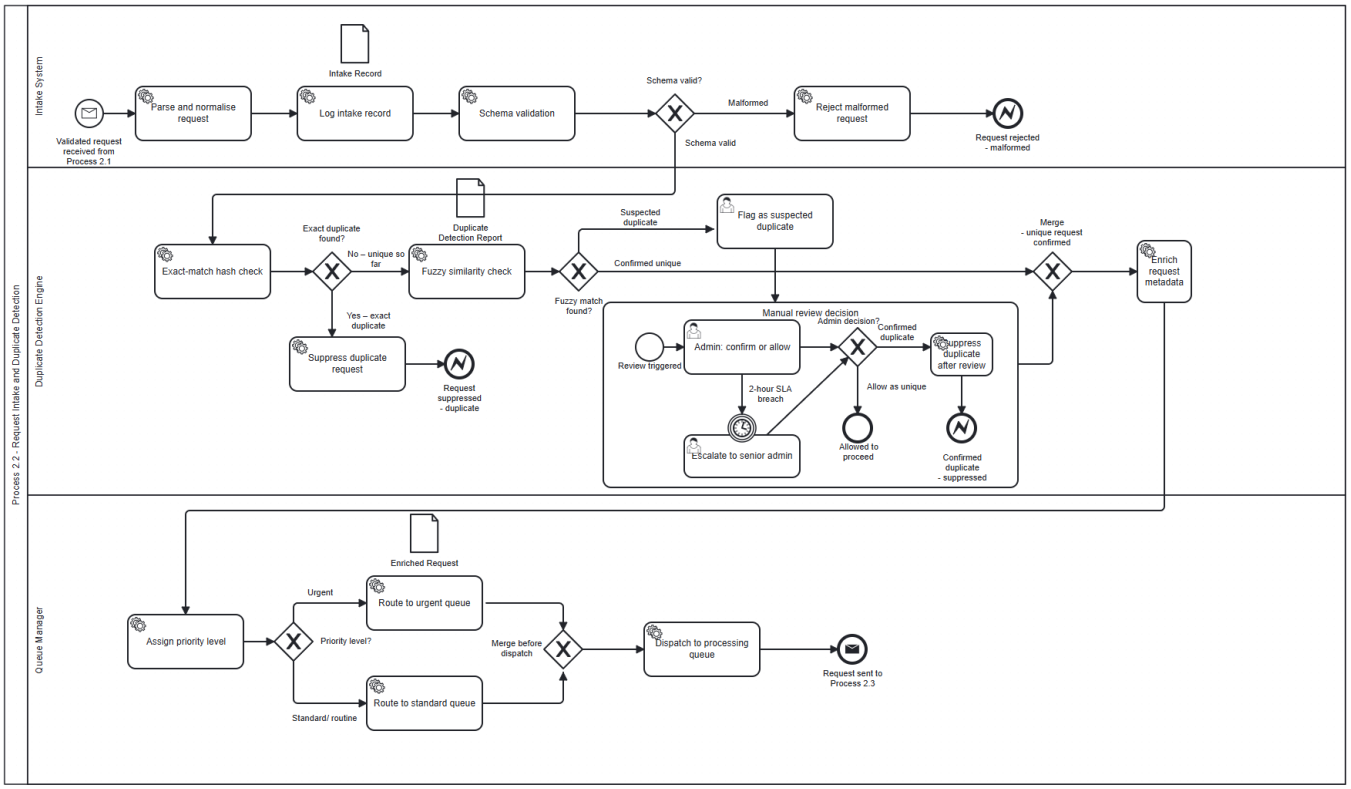
Element	Type	Purpose
Appointment request initiated	Message Start Event	Triggered by incoming patient web or phone request
Select channel	User Task (Patient lane)	Patient chooses web portal or phone submission channel
Fill request form	User Task (Patient lane)	Patient completes appointment request form; outputs Appointment Request data object
Review & submit	User Task (Patient lane)	Patient reviews and confirms the completed form
Receive confirmation	User Task (Patient lane)	Patient receives email or SMS confirmation that the request is queued
Guided form validation	Service Task (Portal System lane)	System performs real-time field validation before submission proceeds
Valid fields?	Exclusive Gateway	Routes to duplicate check if valid; returns to form if errors are found
Flag validation errors	Service Task (Portal System lane)	Highlights invalid fields and loops back to Fill request form
Check duplicate submission	Service Task (Portal System lane)	Queries backend for existing identical requests
Duplicate found?	Exclusive Gateway	Routes to block if duplicate; queues if unique
Notify duplicate & block	Service Task (Portal System lane)	Informs patient of duplicate and blocks new submission
Queue validated request	Service Task (Portal System lane)	Places validated, unique request into processing queue
Queued request received	Intermediate Message Catch Event (Backend lane)	Backend receives queued request from portal via message
Retrieve / register patient record	Service Task (Backend lane)	Looks up patient in DB; initiates registration if not found
Patient known?	Exclusive Gateway	Routes to eligibility check if existing; to record creation if new
Create new patient record	Service Task (Backend lane)	Registers new patient with submitted personal details
Validate patient eligibility	Service Task (Backend lane)	Checks data completeness, format, and eligibility rules
Patient eligible?	Exclusive Gateway	Routes to Process 2.2 if eligible; to rejection if not

Element	Type	Purpose
Forward to Process 2.2	Service Task (Backend lane)	Sends eligible request to Process 2.2 and notifies patient
Reject & notify patient	Service Task (Backend lane)	Rejects request and notifies patient with reason
Appointment Request [Form Data]	Data Object	Represents form data produced by Fill request form task
Request successfully submitted	End Event (plain)	Normal end, request forwarded or rejection sent
Request rejected	Error End Event	Abnormal end for duplicate-blocked or ineligible requests; error code ERR_REQUEST_REJECTED

Assumptions & Constraints

- It is assumed the system supports both a web portal and a phone intake form (used by admin staff on behalf of the patient).
- Phone-based requests are entered by reception staff using the same form. This is modelled as a separate lane within the same pool.
- Duplicate detection is deferred to Process 2.2 to keep this process focused on the patient interaction.

3.2 Process 2.2 - Request Intake & Duplicate Detection



Overview

This back-end process receives the submitted appointment request from Process 2.1, checks for duplicate requests, and forwards valid requests to the patient validation queue. This was not explicitly modelled in the original proposal but is necessary to prevent double submissions, a known pain point.

Key Design Improvements Over Original

- The original proposal mentioned duplicate prevention as a business rule but did not model it as a process step. We make it an explicit service task with a gateway.
- If a duplicate is detected, the patient is notified via message (email/SMS) and the duplicate request is discarded.
- An exclusive gateway routes requests based on duplicate detection results: confirmed unique requests proceed to validation, while suspected duplicates are flagged for manual review by reception staff before being approved or rejected.
- A timer event ensures that if the validation queue does not respond within 5 minutes, an alert is sent to admin staff.

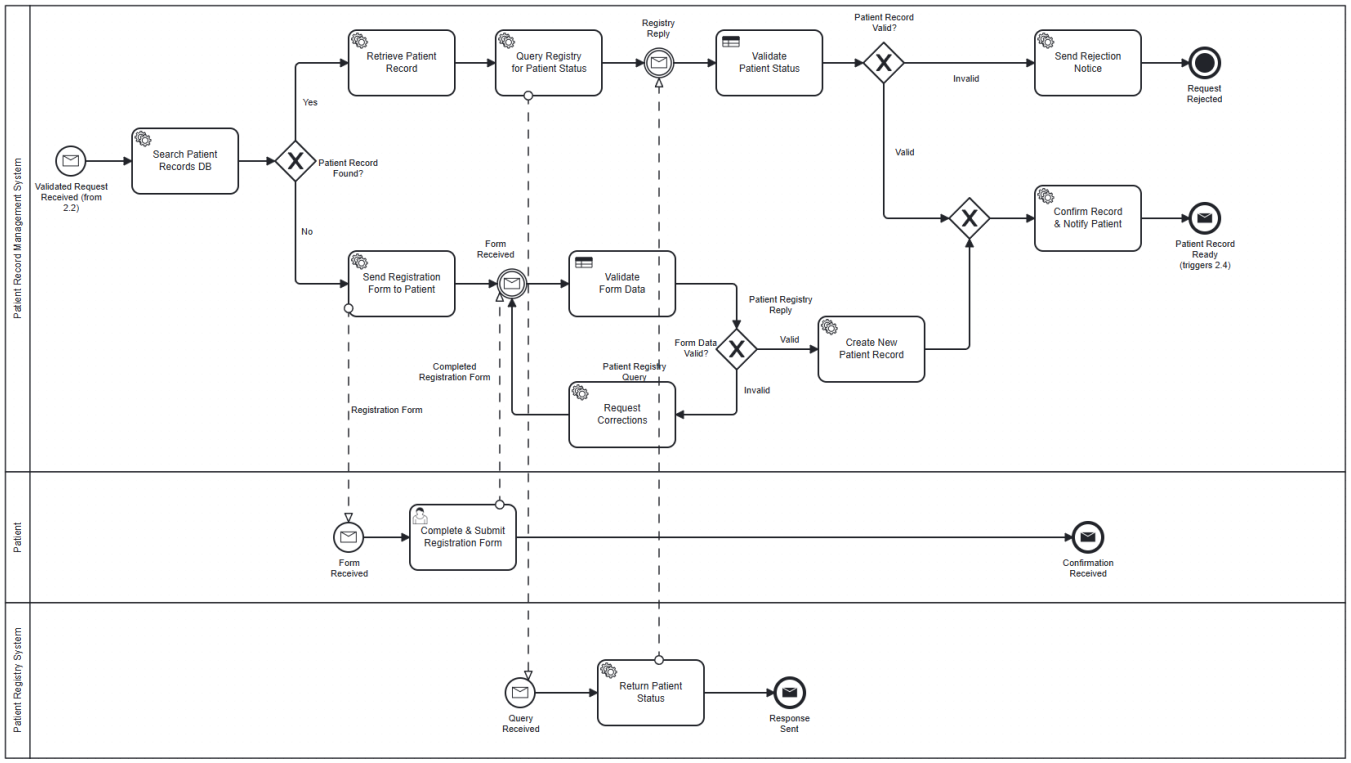
BPMN Elements Used

Element	Type	Purpose
Validated request received from Process 2.1	Message Start Event (Intake System lane)	Triggered by incoming message from Process 2.1

Element	Type	Purpose
Parse and normalise request	Service Task (Intake System lane)	Normalises field formats, strips whitespace, maps to internal schema
Log intake record	Service Task (Intake System lane)	Generates intake ID, timestamps receipt, writes audit entry; outputs Intake Record data object
Schema validation	Service Task (Intake System lane)	Validates required fields and format constraints against business rules
Schema valid?	Exclusive Gateway	Routes to duplicate detection if valid; to rejection if malformed
Reject malformed request	Service Task (Intake System lane)	Logs failure, updates status to REJECTED_MALFORMED, sends error response
Request rejected -- malformed	Error End Event	Terminal state for schema-invalid requests; error code ERR_MALFORMED_REQUEST
Exact-match hash check	Service Task (Duplicate Detection Engine lane)	Computes hash of key fields and checks against last 72h of submissions
Exact duplicate found?	Exclusive Gateway	Routes to suppression if exact match; to fuzzy check if unique
Suppress duplicate and notify	Service Task (Duplicate Detection Engine lane)	Sets status to DUPLICATE_SUPPRESSED and notifies patient with original reference
Request suppressed -- duplicate	Error End Event	Terminal state for exact duplicates; error code ERR_DUPLICATE_REQUEST
Fuzzy similarity check	Service Task (Duplicate Detection Engine lane)	Applies fuzzy matching algorithms; outputs Duplicate Detection Report data object
Fuzzy match found?	Exclusive Gateway	Routes to manual review if suspected duplicate; to enrichment if unique
Flag as suspected duplicate	User Task (Duplicate Detection Engine lane)	Admin reviews suspected duplicate in PENDING_REVIEW hold state
Manual review decision (sub-process)	Expanded Sub-Process	Admin decides CONFIRM_DUPLICATE or ALLOW; includes 2-hour SLA timer for escalation to senior admin
Merge - unique request confirmed	Exclusive Gateway	Convergence point after fuzzy-no or manual-allow paths
Enrich request metadata	Service Task (Duplicate Detection Engine lane)	Adds patient demographics, insurance, urgency score; outputs Enriched Request data object

Element	Type	Purpose
Assign priority level	Service Task (Queue Manager lane)	Applies priority algorithm; outputs URGENT, STANDARD, or ROUTINE label
Priority level?	Exclusive Gateway	Routes to urgent or standard queue lane
Route to urgent queue	Service Task (Queue Manager lane)	Places request in high-priority queue; alerts scheduling staff
Route to standard queue	Service Task (Queue Manager lane)	Places request in standard first-in-first-out queue
Merge before dispatch	Exclusive Gateway	Converges urgent and standard paths before dispatch
Dispatch to processing queue	Service Task (Queue Manager lane)	Writes enriched, prioritised request to queue; fires message to Process 2.3
Request queued, forwarded to Process 2.3	Message End Event	Success terminal state; signals Process 2.3 to begin
Intake Record	Data Object	Produced by Log intake record; audit trail for the request
Duplicate Detection Report	Data Object	Produced by Fuzzy similarity check; records match results
Enriched Request	Data Object	Produced by Enrich metadata; consumed by Dispatch task
2-hour SLA breach (inside sub-process)	Intermediate Timer Catch Event	Escalates to senior admin if no review decision within 2 hours (PT2H)

3.3 Process 2.3 - Patient Record Retrieval & Registration



Overview

This process retrieves the patient's record from the system database. If the patient is not found, a new registration sub-process is triggered. This directly corresponds to the first half of the original Process 2.

Key Design Improvements Over Original

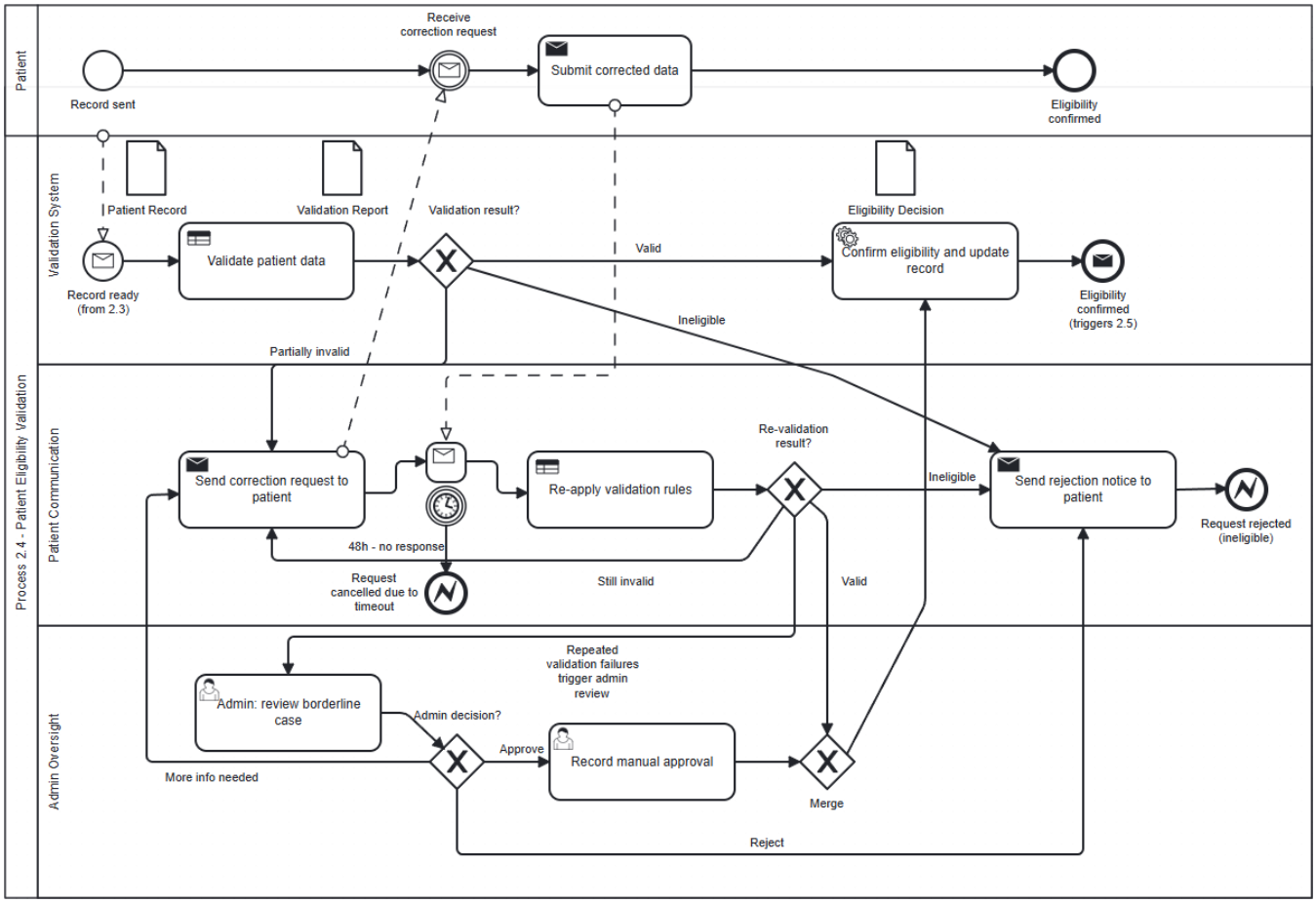
- The original diagram showed 'Register New Patient' as a single task. Our design expands this into a collapsed sub-process containing: capture personal details, assign patient ID, create database record, and send welcome message.
- The Patient Registry Query and Reply are modelled as separate message flows to a dedicated Patient Registry System pool, making the external integration explicit and supporting future failure handling at the system boundary.
- A data store symbol (cylinder) is shown connected to the retrieval task, making the database dependency explicit.
- Lanes are added to show that database retrieval is a system (service) task, while registration assistance may involve a receptionist (user task).
- The registration form sent to new patients during 2.3 is intentionally distinct from the appointment request form in 2.1. Process 2.1 captures only the lightweight details needed to submit a request (name, contact, appointment type, preferred date), while the 2.3 onboarding form collects the extended information required for a full patient record: insurance and PRSI details, GP referral information, medical history declarations, emergency contact, allergies, current medications, and GDPR consent. Existing patients skip this step because their record already contains these fields.

BPMN Elements Used

Element	Type	Purpose
Request Received (System pool)	Message Start Event	Triggered by incoming message from Process 2.2
Search Patient Records DB	Service Task (System pool)	Queries patient records database by submitted details
Record Found?	Exclusive Gateway	Routes to retrieval if record exists; to registration form if new
Retrieve Patient Record	Service Task (System pool)	Retrieves the existing patient record from the database
Query Registry for Patient Status	Service Task (System pool)	Sends patient status query message to Patient Registry System pool
Registry Reply	Intermediate Message Catch Event	Waits for patient status response from Patient Registry pool
Validate Patient Status	Business Rule Task (System pool)	Applies rules to determine if patient eligibility is current and valid
Record Valid?	Exclusive Gateway	Routes to notification if valid; to rejection if invalid
Send Rejection Notice	Service Task (System pool)	Sends rejection notification to patient
Request Rejected	Terminate End Event	Terminates process for invalid/ineligible records
Send Registration Form to Patient	Service Task (System pool)	Sends registration form to Patient pool via message flow
Form Received (System)	Intermediate Message Catch Event	Receives completed registration form from Patient pool; also used as loop-back target for correction requests
Validate Form Data	Business Rule Task (System pool)	Validates submitted registration form fields
Data Valid?	Exclusive Gateway	Routes to record creation if valid; to correction request if invalid
Request Corrections	Service Task (System pool)	Sends correction request back to patient; loops to Form Received
Create New Patient Record	Service Task (System pool)	Creates new patient record in the database
Merge gateway	Exclusive Gateway	Converges existing-record and new-record paths
Confirm Record & Notify Patient	Service Task (System pool)	Confirms record readiness and notifies patient
Record Ready for Process 2.4	Message End Event	Signals that record is ready; triggers Process 2.4

Element	Type	Purpose
Form Received (Patient pool)	Message Start Event	Patient receives registration form sent by System pool
Complete & Submit Registration Form	User Task (Patient pool)	Patient fills and submits the registration form
Confirmation Received (Patient pool)	Message End Event	Patient receives confirmation that process is complete
Query Received (Registry pool)	Message Start Event	Registry receives patient status query from System pool
Return Patient Status	Service Task (Registry pool)	Registry looks up and returns patient status
Response Sent (Registry pool)	Message End Event	Registry end after sending status response
Registration Form (message flow)	Message Flow	Sends form from System pool to Patient pool
Completed Form (message flow)	Message Flow	Returns completed form from Patient pool to System pool
Patient Registry Query (message flow)	Message Flow	Sends query from System pool to Registry pool
Patient Registry Reply (message flow)	Message Flow	Returns status from Registry pool to System pool

3.4 Process 2.4 - Patient Eligibility Validation



Overview

Once the patient record exists in the system, this process validates the data against business rules: field completeness, format correctness (email, phone, DOB), and eligibility criteria. This corresponds to the second half of the original Process 2.

Key Design Improvements Over Original

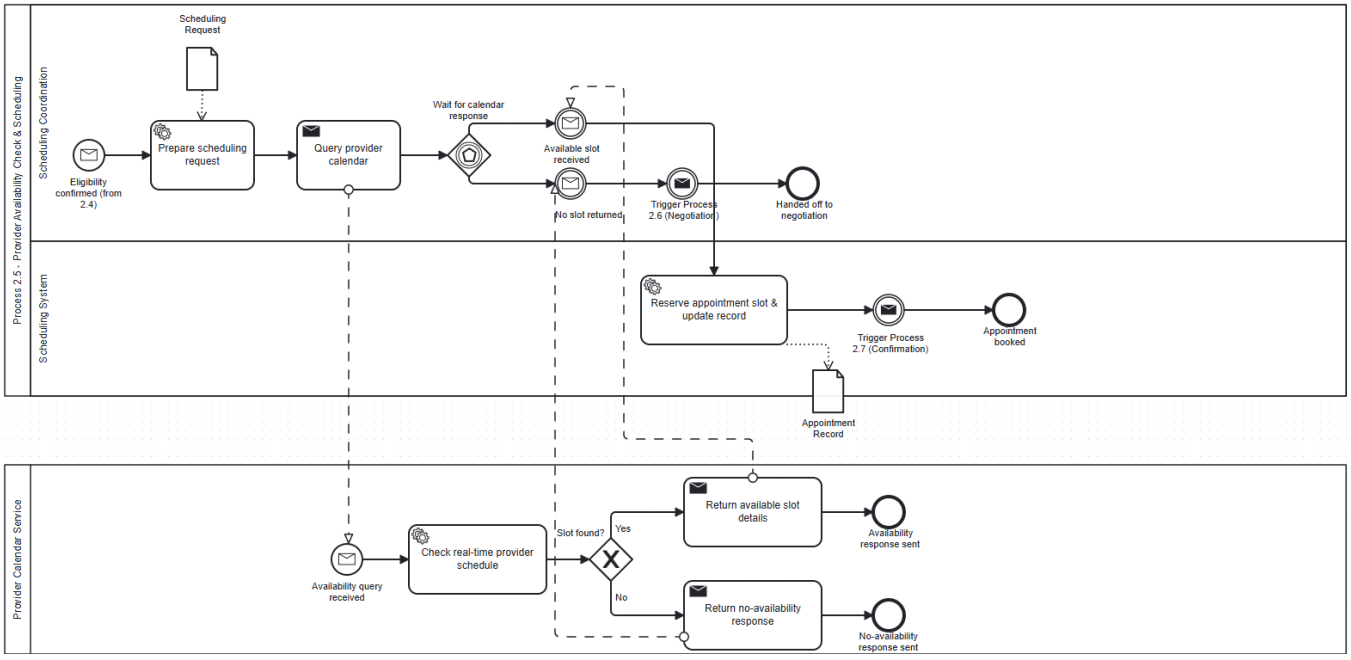
- The original showed a single 'Validate Patient Information' task followed by a gateway. Our design introduces a business rule task that applies all validation rules simultaneously, using a decision table concept.
- Three distinct outcomes are modelled: (1) fully valid, the process proceeds to scheduling; (2) partially invalid, the patient is notified and asked to provide corrections; (3) ineligible, the request is formally rejected with a reason code.
- An intermediate timer catch event on the 'Await Patient Correction' path handles the case where the patient does not respond within 48 hours. The request is then automatically cancelled.
- Message events are used to enable communication between the system and the patient for sending correction requests and receiving updated information.

BPMN Elements Used

Element	Type	Purpose
Record ready (from 2.3)	Message Start Event (Validation System lane)	Triggered by message from Process 2.3 indicating record is ready
Validate patient data	Business Rule Task (Validation System lane)	Applies validation rules; reads Patient Record, outputs Validation Report data object
Validation result?	Exclusive Gateway	Three-way split: Valid = confirm, Partially invalid = notify patient, Ineligible = reject
Confirm eligibility and update record	Service Task (Validation System lane)	Confirms eligibility, updates record; outputs Eligibility Decision data object
Eligibility confirmed (triggers 2.5)	Message End Event	Sends eligibility confirmation to Process 2.5
Send correction request to patient	Send Task (Patient Communication lane)	Sends correction request to Patient pool via message flow
Receive correction (Catch_CorrReceived)	Receive Task (Patient Communication lane)	Waits for corrected data message from Patient pool
Re-apply validation rules	Business Rule Task (Patient Communication lane)	Re-validates corrected patient data
Re-validation result?	Exclusive Gateway	Four-way split: Now valid, Still invalid (retry), Ineligible, Max attempts (escalate)
Send rejection notice to patient	Send Task (Patient Communication lane)	Sends rejection notice for ineligible patients
Request rejected (ineligible)	Error End Event	Terminal rejection state; error code ERR_INELIGIBLE
48h -- no response	Intermediate Timer Event (Patient Communication lane)	Fires if patient does not respond within 48 hours (PT48H)
Timeout -- request cancelled	Error End Event	Terminates on correction timeout; error code ERR_TIMEOUT
Admin: review borderline case	User Task (Admin Oversight lane)	Admin reviews cases that have exceeded maximum correction attempts
Admin decision?	Exclusive Gateway	Three-way: Approve, Reject, or request More info
Record manual approval	User Task (Admin Oversight lane)	Admin records manual approval decision
Merge (GW_MergeApproved)	Exclusive Gateway	Converges re-validated-valid and manually-approved paths
Patient Record	Data Object	Input to business rule validation task

Element	Type	Purpose
Validation Report	Data Object	Output of business rule task; records validation findings
Eligibility Decision	Data Object	Output of confirmation task; records final eligibility outcome
Record sent (Patient pool)	Plain Start Event	Patient pool start, patient waits to receive a correction request
Receive correction request (Patient pool)	Intermediate Message Catch Event	Patient receives correction request from System
Submit corrected data (Patient pool)	Send Task	Patient sends corrected data back to System via message flow
Message flows (correction loop)	Message Flows	MF_CorrReq: System = Patient (correction request); MF_CorrBack: Patient = System (corrected data)

3.5 Process 2.5 - Provider Availability Check & Scheduling



Overview

This process checks real-time provider availability and, if a slot is found at the patient's preferred time, books the appointment. It corresponds to the first path in the original Process 3 ("Provider Available? Yes").

Key Design Improvements Over Original

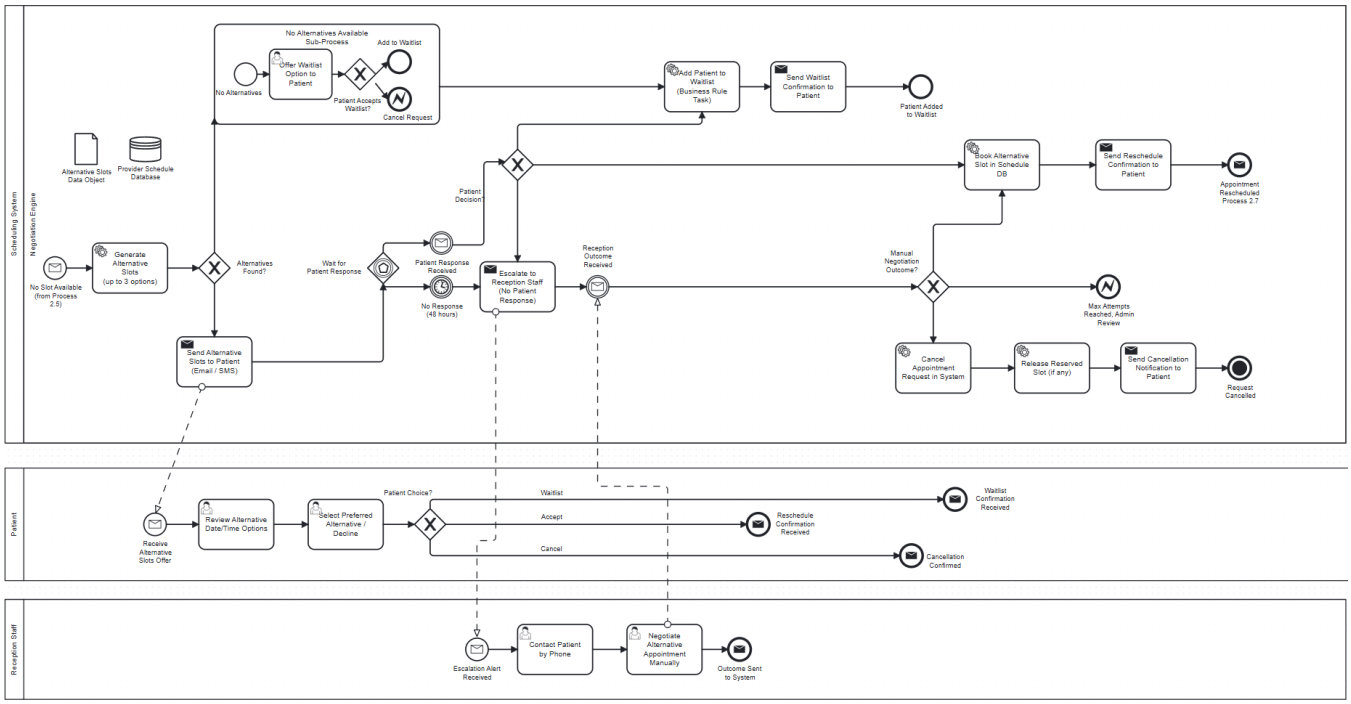
- The original had a basic exclusive gateway for availability. Our design uses an event-based gateway to model a real-world scenario where the system sends a slot query and waits for a response, either an available slot message or a timeout when no slots are found.
- The scheduling task is modelled as a service task with a concurrent data write to the 'Appointment Schedule' data store, ensuring atomicity.
- The scheduling task and the alternative-slot path are explicitly handed off to Process 2.7 and Process 2.6 via message throw events. This separates scheduling logic from communication and negotiation, making each process easier to maintain.
- Collaboration between the Scheduling System pool and the Healthcare Provider pool is shown through message flows, reflecting that provider calendars are treated as an external system.
- The 'Send Appointment Confirmation' step from the original Process 3 top path is removed here and handled in Process 2.7 as a dedicated communication process.

BPMN Elements Used

Element	Type	Purpose
Eligibility confirmed (from 2.4)	Message Start Event (Scheduling Coordination lane)	Triggered by eligibility confirmation message from Process 2.4
Prepare scheduling request	Service Task (Scheduling Coordination lane)	Loads and structures the appointment request for calendar query
Query provider calendar	Send Task (Scheduling Coordination lane)	Sends availability query message to Provider Calendar Service pool
Wait for calendar response	Event-Based Gateway (Scheduling Coordination lane)	Waits for either an available-slot or no-slot message from Calendar pool
Available slot received	Intermediate Message Catch Event	Receives available slot details from Calendar pool
No slot available	Intermediate Message Catch Event	Receives no-availability response from Calendar pool
Reserve appointment slot and update record	Service Task (Scheduling System lane)	Books the slot in the schedule and updates the appointment record
Trigger Process 2.7 (Confirmation)	Intermediate Throw Event (Scheduling System lane)	Throws message event to trigger Process 2.7 (Confirmation & Reminder)
Appointment booked	Plain End Event	Successful scheduling end state
Trigger Process 2.6 (Negotiation)	Intermediate Throw Event (Scheduling Coordination lane)	Throws message event to trigger Process 2.6 (Alternative Scheduling)

Element	Type	Purpose
Handed off to negotiation	Plain End Event	End state when no slot is available; feeds into Process 2.6
Availability query received (Calendar pool)	Message Start Event	Calendar pool receives availability query from Scheduling pool
Check real-time provider schedule	Service Task (Calendar pool)	Queries live provider calendar for matching slots
Slot found? (Calendar pool)	Exclusive Gateway	Routes to send-available or send-unavailable
Return available slot details (Calendar pool)	Send Task	Sends slot details back to Scheduling Coordination via message flow
Return no-availability response (Calendar pool)	Send Task	Sends no-slot response back to Scheduling Coordination via message flow
Message flows (3 total)	Message Flows	System -> Calendar (query), Calendar -> System (available slot), Calendar -> System (no slot)

3.6 Process 2.6 - Alternative Scheduling & Patient Negotiation



Overview

When no provider is available at the patient's preferred time, this process suggests alternatives and negotiates with the patient. It corresponds to the "Provider Available? No" path in the original Process 3.

Key Design Improvements Over Original

- The original modelled 'Suggest Alternative Time or Provider' and 'Review Alternative' as simple sequential tasks. Our design introduces a loop with a counter, with a maximum of three alternatives offered, to prevent infinite loops.
- An intermediate timer catch event attached to the event-based gateway cancels the wait if the patient does not respond within 48 hours. The flow then escalates to reception staff for manual negotiation.
- An event-based gateway is used to wait for either the patient's acceptance message or a rejection message, replacing the simple exclusive gateway in the original model.
- If scheduling is ultimately cancelled, a formal cancellation record is created and the patient receives a cancellation confirmation message.

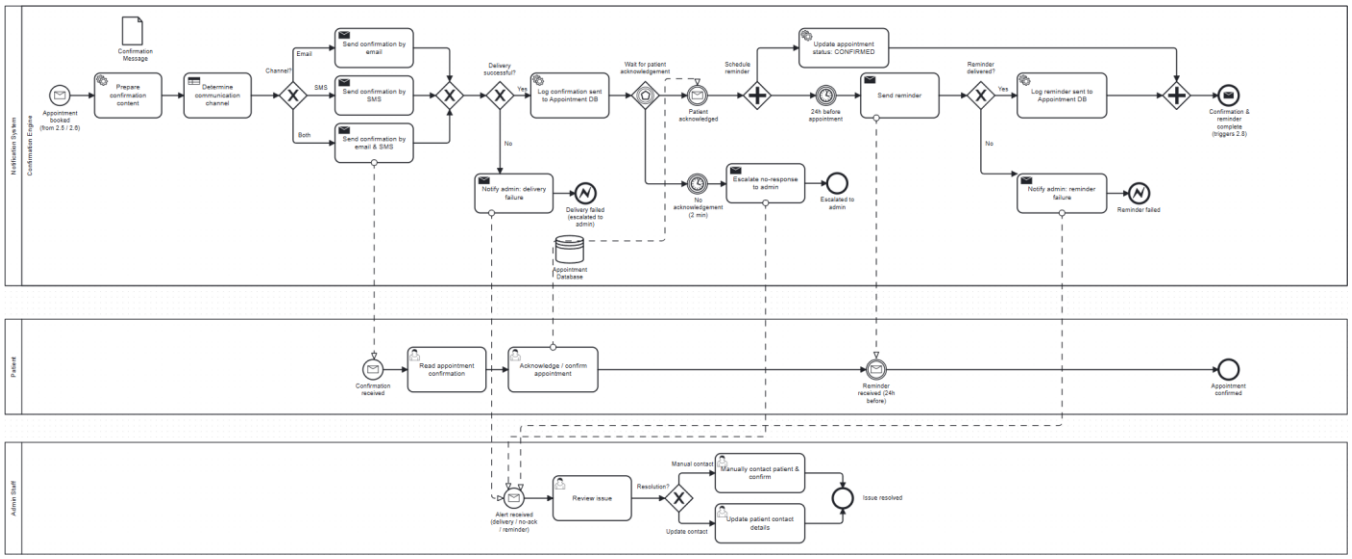
BPMN Elements Used

Element	Type	Purpose
No Slot Available (from Process 2.5)	Message Start Event (Negotiation Engine lane)	Triggered by no-availability message from Process 2.5
Generate Alternative Slots (up to 3)	Service Task (Negotiation Engine lane)	Queries Schedule DB to generate up to 3 alternative slot options
Alternatives Found?	Exclusive Gateway	Routes to send alternatives if found; to No-Alternatives sub-process if none
No Alternatives Available (sub-process)	Expanded Sub-Process	Offers waitlist-only option to patient; patient accepts (add to waitlist) or declines (cancel)
Send Alternative Slots to Patient	Send Task (Negotiation Engine lane)	Sends alternative date/time options to patient via email/SMS (message flow to Patient pool)
Wait for Patient Response	Event-Based Gateway (Negotiation Engine lane)	Waits for patient response message or 48-hour timeout
Patient Response Received	Intermediate Message Catch Event	Receives patient selection response
No Response (48 hours)	Intermediate Timer Catch Event	Fires if no patient response after 48 hours (PT48H)
Patient Decision?	Exclusive Gateway	Three-way: accept alternative, join waitlist, or cancel

Element	Type	Purpose
Book Alternative Slot in Schedule DB	Service Task (Negotiation Engine lane)	Books the accepted alternative slot in the database
Send Reschedule Confirmation to Patient	Send Task (Negotiation Engine lane)	Confirms rescheduled appointment to patient
Appointment Rescheduled = Process 2.7	Message End Event	Successful rescheduling; signals Process 2.7
Add Patient to Waitlist	Service Task (Negotiation Engine lane)	Adds patient to waitlist queue
Send Waitlist Confirmation to Patient	Send Task (Negotiation Engine lane)	Confirms waitlist placement to patient
Patient Added to Waitlist	Plain End Event	End state for waitlist outcome
Escalate to Reception Staff	Send Task (Negotiation Engine lane)	Escalates to Reception Staff pool if timeout or patient cancels
Reception Outcome Received	Intermediate Message Catch Event	Receives negotiation outcome from Reception pool
Manual Negotiation Outcome?	Exclusive Gateway	Three-way: booked, cancelled, or max-attempts-reached
Cancel Appointment Request in System	Service Task (Negotiation Engine lane)	Marks request as cancelled in system
Release Reserved Slot (if any)	Service Task (Negotiation Engine lane)	Releases any held slot back to availability
Send Cancellation Notification to Patient	Send Task (Negotiation Engine lane)	Notifies patient of cancellation
Request Cancelled	Terminate End Event	Hard-stop cancellation end state
Max Attempts Reached, Admin Review	Error End Event	Escalation end when all negotiation options exhausted
Alternative Slots Data Object	Data Object	Holds generated alternative slot options
Provider Schedule Database	Data Store Reference	External data store queried for availability and booked for slot reservation

Element	Type	Purpose
Receive Alternative Slots Offer (Patient pool)	Message Start Event	Patient pool receives alternative offer from Scheduling System
Review Alternative Date/Time Options	User Task (Patient pool)	Patient reviews offered alternatives
Select Preferred Alternative / Decline	User Task (Patient pool)	Patient submits their choice
Escalation Alert Received (Reception pool)	Message Start Event	Reception staff pool receives escalation alert
Contact Patient by Phone	User Task (Reception pool)	Receptionist contacts patient directly
Negotiate Alternative Appointment Manually	User Task (Reception pool)	Receptionist negotiates a slot manually with patient
Message flows (3 cross-pool)	Message Flows	System = Patient (offer), System = Reception (escalation), Reception = System (manual outcome)

3.7 Process 2.7 - Appointment Confirmation & Reminder Dispatch



Overview

This process generates and sends the appointment confirmation to the patient and schedules automated reminder notifications. It directly corresponds to the original Process 4 but is significantly enhanced.

Key Design Improvements Over Original

- The original used a simple exclusive gateway for message delivery. Our design adds an explicit Delivery Successful? gateway. If delivery fails, the system sends an alert to admin staff via a dedicated message flow and ends with an error end event.
- After the patient acknowledges, a parallel gateway splits the flow: one branch updates the appointment status to CONFIRMED, while the other branch waits at an intermediate timer catch event for 24 hours before triggering the reminder send task. This guarantees the reminder fires exactly 24 hours before the appointment, as required by the business rules.
- The confirmation and reminder are modelled as separate parallel paths using a parallel gateway, allowing both processes to run concurrently after confirmation is generated.
- Both delivery failures (initial confirmation and the 24h reminder) terminate at error end events that propagate the failure code, while inline send tasks notify admin staff via message flows. This makes the failure handling visible on the main flow rather than hiding it inside an event sub-process.
- The communication channel, SMS or email, is selected using a business rule task based on the patient's contact preferences stored in their record.

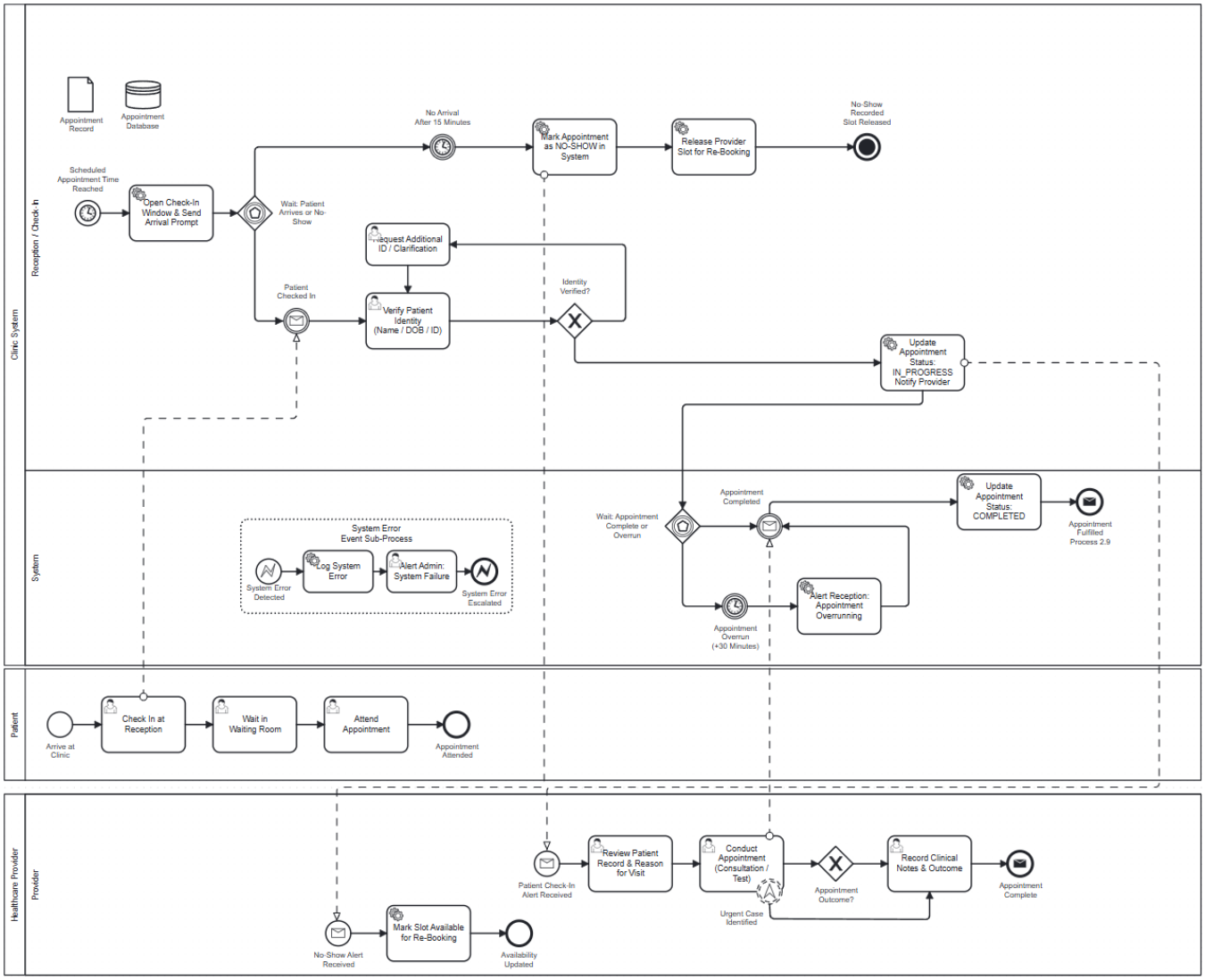
BPMN Elements Used

Element	Type	Purpose
Appointment Booked (from Process 2.5/2.6)	Message Start Event (Confirmation Engine lane)	Triggered by appointment-booked message from Process 2.5 or 2.6
Prepare Confirmation Content	Service Task (Confirmation Engine lane)	Assembles confirmation details: date, time, provider, location
Determine Communication Channel	Business Rule Task (Confirmation Engine lane)	Business rule decides Email, SMS, or both based on patient preferences
Channel selection gateway	Exclusive Gateway	Routes to one of three separate send tasks: Send by Email, Send by SMS, or Send by Email & SMS, based on the channel decision
Send Appointment Confirmation	Send Task (Confirmation Engine lane)	One of three send tasks (Email, SMS, or both) sends the confirmation to the Patient pool via a message flow. All three branches converge at a merge gateway before delivery is checked.
Delivery Successful?	Exclusive Gateway	Routes to log if delivered; triggers Delivery Failure event sub-process if failed
Log Confirmation Sent to Appointment DB	Service Task (Confirmation Engine lane)	Writes confirmation record to Appointment Database
Wait for Patient Acknowledgement	Event-Based Gateway (Confirmation Engine lane)	Waits for patient acknowledgement message or 2-minute timeout

Element	Type	Purpose
Patient Acknowledged	Intermediate Message Catch Event	Receives patient acknowledgement from Patient pool
No Acknowledgement (2 minutes)	Intermediate Timer Catch Event	Fires if no patient ack received within 2 minutes (PT2M)
Escalate No-Response to Admin Staff	Send Task (Confirmation Engine lane)	Escalates to Admin pool if patient does not acknowledge
Escalated to Admin	Plain End Event	End state for no-response escalation path
Send Reminder (Task_SendReminder)	Send Task (Confirmation Engine lane)	Sends the 24h reminder to the Patient pool via message flow after the 24h timer fires
Schedule Reminder (Parallel Split)	Parallel Gateway	Splits flow into two concurrent branches: (1) Update appointment status to CONFIRMED, and (2) wait at 24h timer catch event before sending the reminder. Both branches synchronise at a parallel join gateway.
24h Before Appointment	Intermediate Timer Catch Event	Fires 24 hours before the appointment (PT24H); triggers reminder send
Update Appointment Status: CONFIRMED	Service Task (Confirmation Engine lane)	Sets appointment status to CONFIRMED in database
Reminder Delivered?	Exclusive Gateway	Routes to log reminder if delivered; triggers Reminder Failure sub-process if failed
Log Reminder Sent to Appointment DB	Service Task (Confirmation Engine lane)	Records reminder delivery in Appointment Database
Parallel Join	Parallel Gateway	Synchronises reminder-log and status-update paths
Confirmation & Reminder Complete = Process 2.8	Message End Event	Signals Process 2.8 that confirmation and reminder cycle is complete
Notify Admin: Delivery Failure	Send Task (Confirmation Engine lane)	Notifies admin of delivery failure via message flow; followed by an error end event
Notify Admin: Reminder Failure	Send Task (Confirmation Engine lane)	Notifies admin of reminder failure via message flow; followed by an error end event
Appointment Database	Data Store Reference	Stores confirmation and reminder log entries
Confirmation Message Object	Data Object	Holds assembled confirmation content

Element	Type	Purpose
Read Appointment Confirmation (Patient pool)	User Task	Patient reads the confirmation message
Acknowledge / Confirm Appointment (Patient pool)	User Task	Patient sends acknowledgement back to Notification System
Reminder Received (Patient pool)	Intermediate Message Catch Event	Patient receives 24h reminder message
Alert Received (Admin pool)	Message Start Event	Admin staff receive delivery-failure or no-response alert
Review Issue (Admin pool)	User Task	Admin reviews the delivery or response issue
Manually Contact Patient & Confirm (Admin pool)	User Task	Admin manually contacts patient to confirm appointment
Message flows (6 cross-pool)	Message Flows	System -> Patient (confirmation, reminder), Patient -> System (acknowledgement), System -> Admin (delivery failure, no-acknowledgement, reminder failure)

3.8 Process 2.8 - Appointment Fulfilment



Overview

This process covers the day of the appointment, including patient check-in, identity verification, and the conduct of the appointment itself. It corresponds to the first half of the original Process 5.

Key Design Improvements Over Original

- The original showed a simple linear sequence. Our design adds a no-show handling path. If the patient has not checked in within 15 minutes of the scheduled time, based on the business rules, the appointment is marked as 'No Show' using a timer boundary event.
- Identity verification is modelled as a user task with a boundary error event. If verification fails, for example if the wrong patient presents, the process escalates to a security review sub-process.
- A collaboration diagram is used to show interaction between three pools: Patient, Reception Staff, and Healthcare Provider, making responsibilities clear.
- A message flow is used to notify the Healthcare Provider pool that the patient has been verified and the consultation can begin.

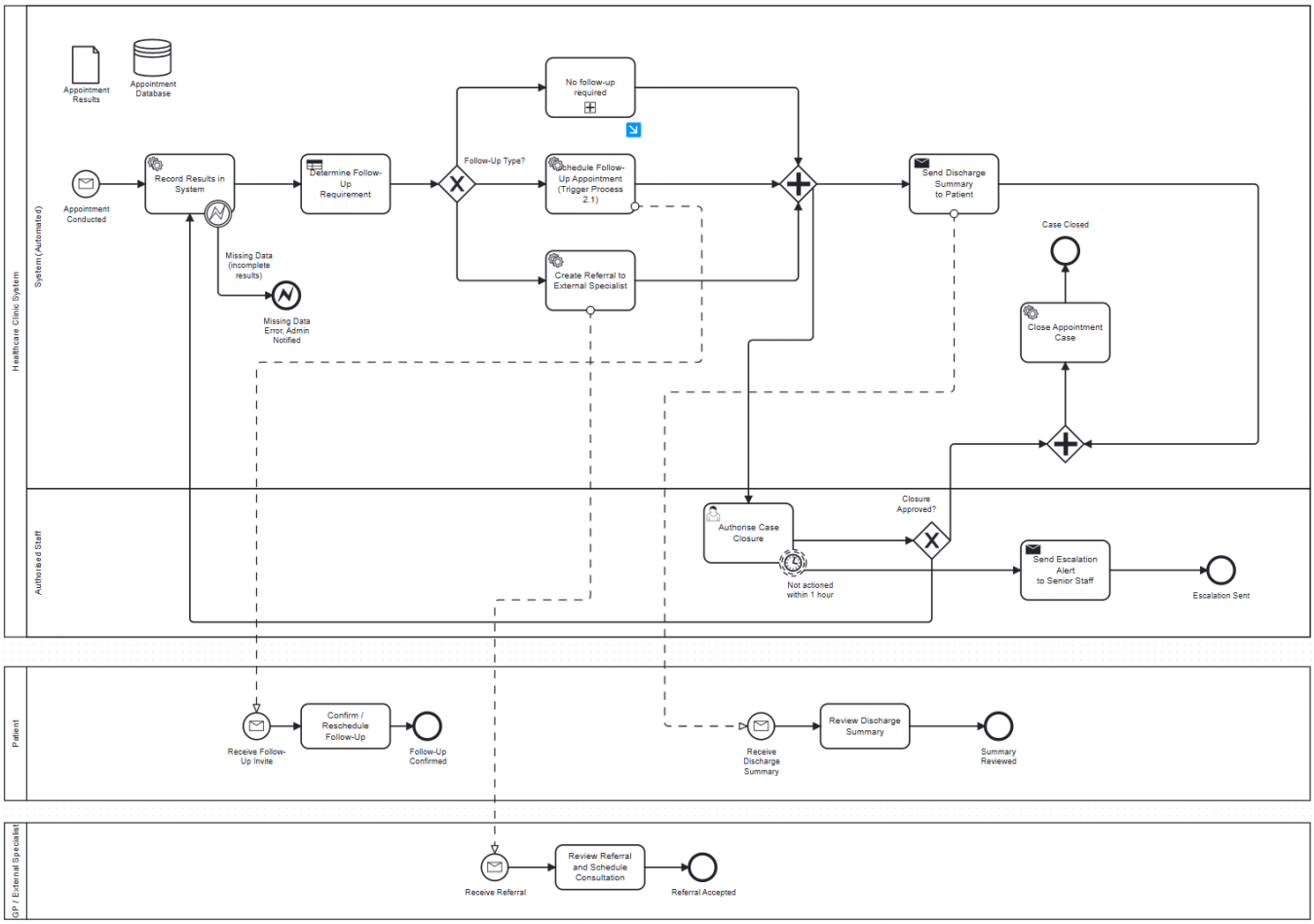
BPMN Elements Used

Element	Type	Purpose
Scheduled Appointment Time Reached	Timer Start Event (Reception/Check-In lane)	Fires at the scheduled appointment date/time (timeDate: appointmentDateTime)
Open Check-In Window & Send Arrival Prompt	Service Task (Reception/Check-In lane)	Opens the check-in window and sends arrival prompt to patient
Wait: Patient Arrives or No-Show	Event-Based Gateway (Reception/Check-In lane)	Waits for patient check-in message or 15-minute no-show timer
Patient Checked In	Intermediate Message Catch Event	Receives check-in message from Patient pool
No Arrival After 15 Minutes	Intermediate Timer Catch Event	Fires if no patient arrival within 15 minutes (PT15M)
Mark Appointment as NO-SHOW in System	Service Task (Reception/Check-In lane)	Updates appointment status to NO_SHOW; sends alert to Provider pool
Release Provider Slot for Re-Booking	Service Task (Reception/Check-In lane)	Releases the slot so it can be re-booked
No-Show Recorded -> Slot Released	Terminate End Event	Hard-stop end for no-show path
Verify Patient Identity (Name/DOB/ID)	User Task (Reception/Check-In lane)	Receptionist verifies patient identity
Identity Verified?	Exclusive Gateway	Routes to update status if verified; loops to Request Additional ID if not
Request Additional ID / Clarification	User Task (Reception/Check-In lane)	Receptionist requests further identification from patient; loops back to verify
Update Appointment Status: IN_PROGRESS = Notify Provider	Service Task (Reception/Check-In lane)	Sets status to IN_PROGRESS; sends check-in alert to Provider pool via message flow
Wait: Appointment Complete or Overrun	Event-Based Gateway (System lane)	Waits for completion message from Provider or 30-minute overrun timer
Appointment Completed	Intermediate Message Catch Event (System lane)	Receives appointment-complete signal from Healthcare Provider pool
Appointment Overrun (+30 Minutes)	Intermediate Timer Catch Event (System lane)	Fires if appointment has not completed within 30 minutes (PT30M)

Element	Type	Purpose
Alert Reception: Appointment Overrunning	Service Task (System lane)	Notifies reception of overrun; then rejoins completion path
Update Appointment Status: COMPLETED	Service Task (System lane)	Sets appointment status to COMPLETED in database
Appointment Fulfilled = Process 2.9	Message End Event	Signals Process 2.9 that appointment is complete
System Error Event Sub-Process	Event Sub-Process (triggered by error)	Catches system error: logs error, alerts admin via user task
Appointment Record	Data Object	Represents the appointment record used during check-in
Appointment Database	Data Store Reference	Stores appointment status updates
Arrive at Clinic (Patient pool)	Plain Start Event	Patient arrives at clinic
Check In at Reception (Patient pool)	User Task	Patient performs check-in; sends message to Clinic System
Wait in Waiting Room (Patient pool)	User Task	Patient waits to be called
Attend Appointment (Patient pool)	User Task	Patient participates in the appointment
Patient Check-In Alert Received (Provider pool)	Message Start Event	Provider receives check-in notification from Clinic System
Review Patient Record & Reason for Visit	User Task (Provider pool)	Provider reviews records before conducting appointment
Conduct Appointment (Consultation/Test)	User Task (Provider pool)	Provider conducts the appointment; has non-interrupting boundary event for urgent cases
Urgent Case Identified	Non-Interrupting Boundary Escalation Event	Fires if urgent case is identified during consultation; routes to Record Notes without stopping appointment
Appointment Outcome?	Exclusive Gateway (Provider pool)	Routes to Record Clinical Notes (standard outcome)
Record Clinical Notes & Outcome	User Task (Provider pool)	Provider records notes; end event sends completion signal to Clinic System
No-Show Alert Received (Provider pool)	Message Start Event	Provider receives no-show alert from Clinic System
Mark Slot Available for Re-Booking (Provider pool)	Service Task (Provider pool)	Provider marks their slot as available again

Element	Type	Purpose
Message flows (4 cross-pool)	Message Flows	Patient = System (check-in), System = Provider (check-in alert), System = Provider (no-show alert), Provider = System (appointment done)

3.9 Process 2.9 - Post-Appointment Processing & Case Closure



Overview

After the appointment is conducted, this process records results, determines if a follow-up is needed, schedules any follow-ups, and closes the case. It corresponds to the second half of the original Process 5.

Key Design Improvements Over Original

- The original showed a simple exclusive gateway, "Follow-up Required?". Our design introduces a parallel path where results are recorded in the system while a discharge summary is sent to the patient's GP, if applicable.

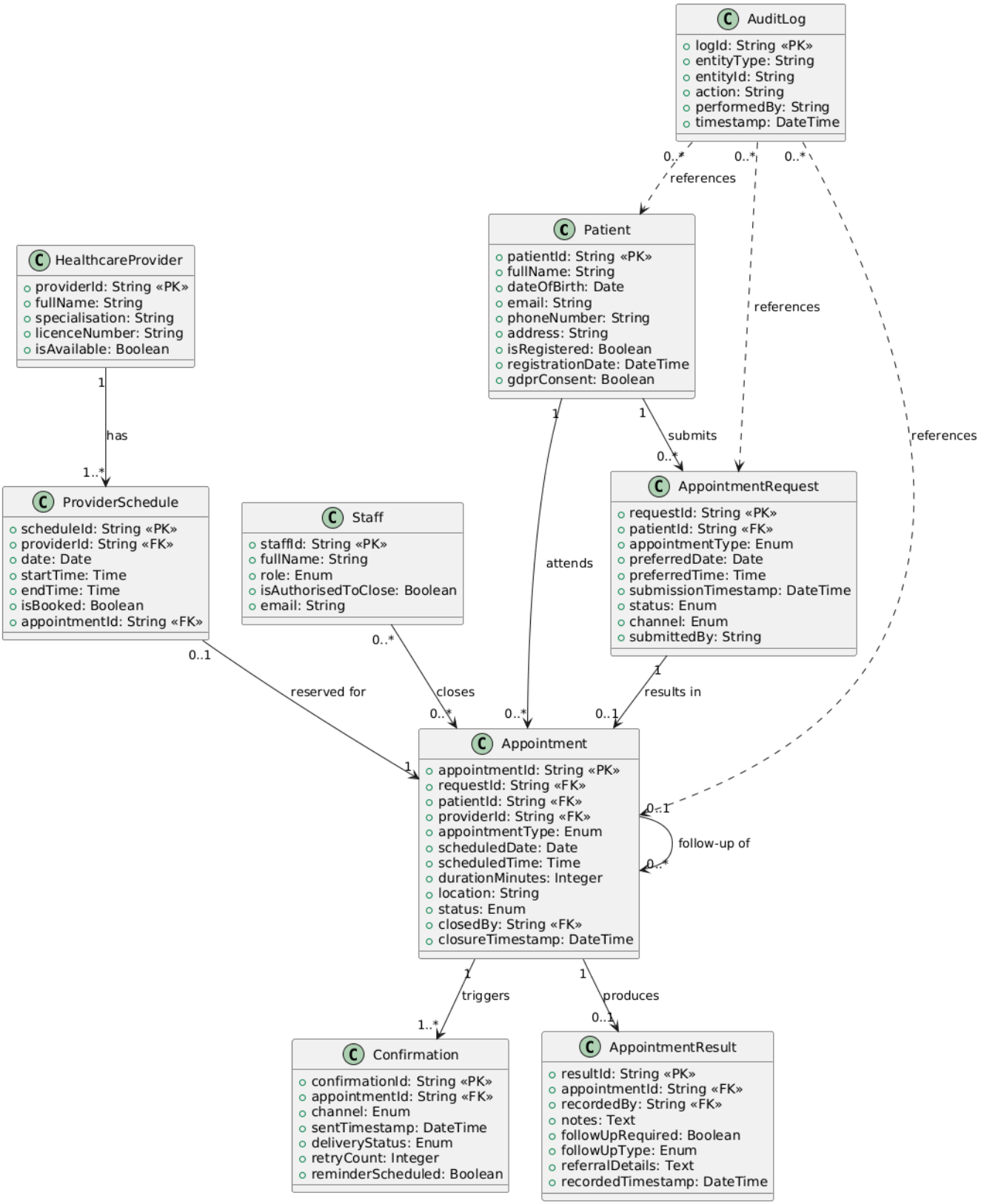
- A business rule task is used to determine the follow-up type based on appointment results, replacing the simple yes or no gateway.
- Authorisation for case closure is modelled as a user task that requires approval from authorised staff, based on the business rules, rather than being an automatic step.
- A data object, 'Appointment Results', is attached to the recording task and flows into both the follow-up scheduling and case closure tasks.
- An error end event is used if the case cannot be closed due to missing data. In this case, an admin notification is triggered.

BPMN Elements Used

Element	Type	Purpose
Appointment Conducted	Message Start Event (System lane)	Triggered by message from Process 2.8 indicating appointment is complete
Record Results in System	Service Task (System lane)	Writes outcomes, diagnosis, and treatment to Appointment DB; reads Appointment Results data object
Missing Data (incomplete results)	Error Boundary Event on Record Results	Fires if results are incomplete; routes to error end event
Missing Data Error -- Admin Notified	Error End Event	Escalation end when results data is missing; error code ERR_MISSING_DATA
Determine Follow-Up Requirement	Business Rule Task (System lane)	Applies clinical decision table to determine follow-up type: None, Internal or External Referral
Follow-Up Type?	Exclusive Gateway	Three-way split: No Follow-Up = sub-process, Internal Follow-Up = trigger task, External Referral = referral task
No Follow-Up Path	Collapsed Sub-Process	Updates appointment status to Completed with no further action; output feeds Parallel Split
Schedule Follow-Up Appointment (Trigger Process 2.1)	Service Task (System lane)	Creates pre-populated AppointmentRequest and sends follow-up invite to Patient pool via message flow
Create Referral to External Specialist	Service Task (System lane)	Generates referral document and sends to GP/Specialist pool via message flow
Parallel Split	Parallel Gateway	Simultaneously sends discharge summary to patient and routes to authorisation for case closure
Send Discharge Summary to Patient	Send Task (System lane)	Sends appointment outcome summary (diagnosis, treatment, prescriptions) to Patient pool
Parallel Join	Parallel Gateway	Synchronises discharge-sent and closure-approved paths before closing case

Element	Type	Purpose
Authorise Case Closure	User Task (Authorised Staff lane)	Authorised staff reviews record and approves closure; has non-interrupting 1-hour timer boundary event
Not actioned within 1 hour	Non-Interrupting Timer Boundary Event on Authorise Closure	Fires if closure is not authorised within 1 hour (PT1H); triggers escalation alert
Send Escalation Alert to Senior Staff	Send Task (Authorised Staff lane)	Notifies senior staff that closure is overdue
Escalation Sent	Plain End Event	End state for escalation path
Closure Approved?	Exclusive Gateway (Authorised Staff lane)	Approved = Parallel Join; Rejected = loops back to Record Results
Close Appointment Case	Service Task (System lane)	Sets status to COMPLETED, archives record, updates ProviderSchedule, writes audit log entry
Case Closed	Plain End Event	Normal process end, appointment archived as COMPLETED
Appointment Results	Data Object	Input to Record Results task; holds raw appointment outcome data
Appointment Database	Data Store Reference	Stores all written appointment results and closure records
Receive Discharge Summary (Patient pool)	Message Start Event	Patient receives discharge summary from Clinic System
Review Discharge Summary (Patient pool)	Task	Patient reads the discharge summary
Receive Follow-Up Invite (Patient pool)	Message Start Event	Patient receives follow-up appointment invite
Confirm / Reschedule Follow-Up (Patient pool)	Task	Patient confirms or reschedules the follow-up
Receive Referral (GP/Specialist pool)	Message Start Event	GP/Specialist receives referral document from Clinic System
Review Referral and Schedule Consultation (GP pool)	Task	GP/Specialist reviews referral and arranges consultation
Message flows (3 cross-pool)	Message Flows	System = Patient (discharge summary), System = Patient (follow-up invite), System = GP (referral)

4. System Design - UML Class Diagram



The UML Class Diagram below represents the data structures required to support all nine processes. The classes are derived directly from the entities referenced in the BPMN models and the business rules defined by the client.

4.1 Class Descriptions

Patient

Central entity representing a person registered in the system. Patients may have zero or many appointments.

Attribute	Type	Notes
patientId	String (PK)	Unique system-generated ID
fullName	String	Required full legal name
dateOfBirth	Date	Must be a past date
email	String	Must follow valid email format
phoneNumber	String	Must be a valid Irish number
address	String	Optional
isRegistered	Boolean	True once registration is complete
registrationDate	DateTime	Timestamp of registration
gdprConsent	Boolean	GDPR consent flag, required before data storage

AppointmentRequest

Represents a patient's submission before it is confirmed as a scheduled appointment. Exists in a pending/validation state.

Attribute	Type	Notes
requestId	String (PK)	Unique request identifier
patientId	String (FK)	Links to Patient
appointmentType	Enum	CONSULTATION | TEST | FOLLOW_UP
preferredDate	Date	Patient's preferred appointment date
preferredTime	Time	Patient's preferred time slot
submissionTimestamp	DateTime	When the request was submitted
status	Enum	PENDING | VALIDATED | SCHEDULED | DUPLICATE | REJECTED
channel	Enum	ONLINE | PHONE
submittedBy	String	PatientId or StaffId (for phone submissions)

Appointment

A confirmed, scheduled appointment. Created from a validated AppointmentRequest once a provider slot is booked.

Attribute	Type	Notes
appointmentId	String (PK)	Unique appointment identifier
requestId	String (FK)	Links to originating request
patientId	String (FK)	Links to Patient
providerId	String (FK)	Links to HealthcareProvider
appointmentType	Enum	Inherits from request
scheduledDate	Date	Confirmed date
scheduledTime	Time	Confirmed start time
durationMinutes	Integer	Predefined by appointment type
location	String	Clinic room or telehealth link
status	Enum	SCHEDULED | CONFIRMED | COMPLETED | CANCELLED | NO_SHOW
closedBy	String (FK)	StaffId of authorised closer
closureTimestamp	DateTime	When case was closed

HealthcareProvider

Represents a doctor, nurse, or specialist available in the system.

Attribute	Type	Notes
providerId	String (PK)	Unique provider identifier
fullName	String	Provider's full name
specialisation	String	Medical specialisation/role
licenceNumber	String	Professional registration number
isAvailable	Boolean	Current availability flag

ProviderSchedule

Represents a time slot in a provider's calendar. Links providers to available or booked times.

Attribute	Type	Notes
scheduleId	String (PK)	Unique schedule entry
providerId	String (FK)	Links to HealthcareProvider
date	Date	Date of the slot
startTime	Time	Start of the slot

Attribute	Type	Notes
endTime	Time	End of the slot
isBooked	Boolean	True if slot is reserved
appointmentId	String (FK, nullable)	Linked appointment if booked

Confirmation

Records the confirmation message sent to a patient. Tracks delivery status and retry attempts.

Attribute	Type	Notes
confirmationId	String (PK)	Unique confirmation record
appointmentId	String (FK)	Links to Appointment
channel	Enum	EMAIL , SMS , PHONE
sentTimestamp	DateTime	When confirmation was dispatched
deliveryStatus	Enum	SENT | DELIVERED | FAILED | RETRY
retryCount	Integer	Number of delivery retries (max 1)
reminderScheduled	Boolean	Whether a 24hr reminder has been queued

AppointmentResult

Records the outcome of a completed appointment including clinical notes and follow-up determination.

Attribute	Type	Notes
resultId	String (PK)	Unique result record
appointmentId	String (FK)	Links to Appointment
recordedBy	String (FK)	ProviderId who recorded the result
notes	Text	Clinical notes (access controlled)
followUpRequired	Boolean	Whether a follow-up is needed
followUpType	Enum	NONE | INTERNAL | REFERRAL
referralDetails	Text	Details if referral is required
recordedTimestamp	DateTime	When result was recorded

Staff

Represents administrative and scheduling staff who interact with the system.

Attribute	Type	Notes
staffId	String (PK)	Unique staff identifier

Attribute	Type	Notes
fullName	String	Staff member's name
role	Enum	RECEPTIONIST , SCHEDULER , ADMIN , PROVIDER
isAuthorisedToClose	Boolean	Whether staff can close appointments
email	String	Staff email for system notifications

AuditLog

Tracks all system events for GDPR compliance and operational tracing.

Attribute	Type	Notes
logId	String (PK)	Unique log entry
entityType	String	Eg., 'Appointment', 'Patient'
entityId	String	ID of the affected entity
action	String	Eg., 'CREATED', 'UPDATED', 'CLOSED'
performedBy	String	StaffId or 'SYSTEM'
timestamp	DateTime	When the action occurred

4.2 Key Relationships

From Class	To Class	Multiplicity	Relationship
Patient	AppointmentRequest	1 to 0..*	A patient may have many requests
Patient	Appointment	1 to 0..*	A patient may have many confirmed appointments
AppointmentRequest	Appointment	1 to 0..1	A request results in at most one appointment
HealthcareProvider	ProviderSchedule	1 to 1..*	A provider has many schedule slots
ProviderSchedule	Appointment	0..1 to 1	A booked slot maps to one appointment
Appointment	Confirmation	1 to 1..*	An appointment may have multiple confirmation attempts
Appointment	AppointmentResult	1 to 0..1	A completed appointment has one result record
Appointment	Appointment (follow-up)	0..1 to 0..*	Follow-up appointments self-reference
Staff	Appointment	0..* to 0..*	Staff can close many appointments

From Class	To Class	Multiplicity	Relationship
AuditLog	All Entities	0..* to 1	Audit log references any entity

5. Discussion

5.1 Design Decisions & Improvements

Our design makes several significant improvements over the original client proposal:

1. **Process Decomposition:** The original five processes contained complexity that made them difficult to model using advanced BPMN elements. Decomposing them into nine processes allowed us to apply appropriate element types at each stage, including message events at boundaries, timer events for SLAs, and error events for failures.
2. **Advanced BPMN Elements:** The original models used only basic elements such as start and end events, tasks, and exclusive gateways. Our design incorporates timer events, error events, message events, event-based gateways, parallel gateways, send and receive tasks, business rule tasks, sub-processes, boundary events, terminate events, and full multi-pool collaboration with message flows, all of which are required by the project specification.
3. **SLA Enforcement:** The client's non-functional requirements specified that requests should be processed within tight time windows and confirmations sent within minutes. These SLAs are embedded as timer catch events in Processes 2.4, 2.6, 2.7, and 2.8, making them operational and enforceable rather than just stated requirements.
4. **GDPR Compliance:** The client mentioned GDPR but did not model it. Our design includes a GDPR consent attribute in the Patient class and incorporates it as a validation check in Process 2.4. The AuditLog class ensures that all data access is traceable.
5. **Error Handling:** The original proposal identified pain points such as incomplete data, delivery failures, and no-shows but did not model exception paths. Our design introduces explicit error handling for each of these scenarios.

5.2 Assumptions Made

- The system is assumed to have a RESTful integration with provider calendar systems, such as a scheduling API. This is why availability queries are modelled as service tasks using event-based gateways.
- The communication gateway, including SMS and email services, is assumed to be an external third-party system with its own availability SLA. This justifies the use of an error event sub-process in Process 2.7.
- Phone-based appointment requests are assumed to be entered by reception staff using the same digital form as the online portal. A separate phone-only workflow is therefore not required.
- The system is assumed to operate during clinic operating hours, Monday to Friday from 08:00 to 18:00 Irish time. Timer events are assumed to respect these hours.
- Follow-up appointments are assumed to restart the appointment request process, Process 2.1, rather than being directly booked. This ensures they go through the same validation pipeline.

5.3 Constraints & Clarifications

- **Urgent appointments:** The business rules mention that urgent appointments may override standard scheduling rules. This is noted as a constraint, but the override path is not modelled because the conditions for urgency were not defined by the client. Further clarification would be required.

- **Walk-in appointments:** The client stated that walk-ins are generally accepted only in urgent cases. This scenario is not included in the main process and would require an event sub-process triggered by a staff-initiated signal event.
- **Provider pool structure:** It is unclear whether each provider manages their own calendar or if scheduling is centralised. We have assumed a centralised scheduling system supported by provider APIs.
- **Cancellation by patient:** The client did not include patient-initiated cancellations. This is a significant missing process and should be added in future iterations.

5.4 Missing Elements from Original Proposal

Missing Element	Impact	Our Design Decision
No exception handling for system failures	High - system downtime would halt all bookings	Added error end events and admin escalation paths through dedicated send tasks in affected processes
No patient-initiated cancellation process	High - patients cannot cancel confirmed appointments	Noted as out-of-scope; recommended as a future process (Process 2.10)
No no-show re-booking path	Medium - system does not recover appointment slots	Added slot release service task in Process 2.8 no-show path
No GDPR consent capture	High - legal requirement in Ireland/EU	Added to Patient registration sub-process and data model
No staff authentication	High - security requirement	Added role-based access as a non-functional design note; modelled as a condition on the 'Authorise Case Closure' user task
No audit trail	Medium - required for regulatory compliance	Added AuditLog class to UML diagram; linked to all write operations

6. Conclusion

This report presents a comprehensive system analysis and design for the Healthcare Appointment Management System proposed by Group 11:11. Our design transforms five high-level business processes into nine logically structured, implementation-ready processes modelled with advanced BPMN elements.

Key achievements of our design include:

- Full process decomposition, enabling clear responsibility assignment across patient, reception, scheduling, and provider pools
- Comprehensive use of advanced BPMN elements, including timer events, error events, message events, event-based gateways, parallel gateways, business rule tasks, send and receive tasks, sub-processes, boundary events, and multi-pool collaboration diagrams
- A complete UML Class Diagram with nine interconnected entities, their attributes, and relationships, derived directly from the process models
- Explicit SLA enforcement through timer catch events aligned with the client's non-functional requirements
- Systematic identification and resolution of missing elements from the original proposal, including GDPR compliance, error handling, no-show management, and audit logging

The resulting design provides a solid foundation for implementation, with sufficient detail for developers to begin building the system while remaining aligned with the client group's original intent and the real-world operational context of Irish healthcare appointment management.

References & Appendix

A. Process Landscape Quick Reference

Process	Input From	Output To	Key BPMN Elements
2.1 Patient Request Portal	Patient	Process 2.2	Message Start, Timer Catch Events, Error End Events, Sub-Process, User Tasks, Message End
2.2 Request Intake	Process 2.1	Process 2.3	Message Start, Service Tasks, User Tasks, Exclusive Gateways, Terminate End, Multi-Pool Collaboration
2.3 Record Retrieval	Process 2.2	Process 2.4	Message Start, Business Rule Task, Service Tasks, Data Objects, Timer Catch Events, Error End Events, Multi-Pool Collaboration
2.4 Eligibility Validation	Process 2.3	Process 2.5	Message Events, Event-Based Gateway, Send Tasks, Service Tasks, Data Objects, Multi-Pool Collaboration
2.5 Availability & Scheduling	Process 2.4	Process 2.7	Event-Based Gateway, Message/Timer Events, Service Task, Data Store

Process	Input From	Output To	Key BPMN Elements
2.6 Alternative Scheduling	Process 2.5	Process 2.7 or End	Event-Based Gateway, Timer Catch Event, Sub-Process, Send Tasks, Terminate End, Error End, Multi-Pool Collaboration
2.7 Confirmation & Reminders	Process 2.5/2.6	Patient	Parallel Gateway, Business Rule Task, Send Task, Timer Start, Error Sub-Process
2.8 Appointment Fulfilment	Timer (scheduled time)	Process 2.9	Timer Catch Events, Boundary Events, Parallel Gateways, User Tasks, Sub-Process, Collaboration
2.9 Post-Appointment Closure	Process 2.8	End / Process 2.1	Business Rule Task, Parallel Gateway, Data Object, User Task, Error End

B. Business Rules Reference

The following business rules from the client proposal are reflected in our process and data designs:

Rule Category	Rule	Implemented In
Patient Data	Mandatory fields: name, DOB, contact, appointment type	Process 2.1 (validation gateway), Process 2.4 (business rule task)
Patient Data	Email, phone, DOB format validation	Process 2.4 (business rule task), Patient class attributes
Patient Data	No duplicate requests (same patient, date, time)	Process 2.2 (service task + gateway)
Patient Data	Patient must exist before appointment confirmed	Process 2.3 (existence check) and FK in Appointment class
Scheduling	Only during operating hours, provider available	Process 2.5 (availability query + timer)
Scheduling	No double booking of a provider	ProviderSchedule.isBooked flag; checked in Process 2.5
Scheduling	Slot duration matches appointment type	AppointmentType is mapped to duration in the service task
Scheduling	Suggest alternatives if unavailable	Process 2.6 (full alternative negotiation process)
Communication	Confirmation on approval; includes date/time/location/provider	Process 2.7 Confirmation class
Communication	Reminder 24 hours before	Process 2.7 (parallel gateway with 24h timer catch event before reminder send)
Communication	Invalid contact = notify admin	Process 2.7 (delivery failure path)
Post-Appointment	Status changes on appointment time	Process 2.8 (timer start), Appointment.status enum

Rule Category	Rule	Implemented In
Post-Appointment	No-show after 15 minutes	Process 2.8 (timer boundary event, 15 min)
Post-Appointment	Only authorised staff can close	Process 2.9 (user task with role check), Staff.isAuthorisedToClose